

Characterizing the Flaws of Image-Based AI-Generated Content

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Abstract

The advancement of foundation models provides opportunities to efficiently generate multimodal content with reduced manual labor and time. Such AI-generated content (AIGC), however, can often be inaccurate, misleading, surreal, or hallucinating. Researchers developed multiple ways to systemize our understanding of the bias, errors, and failures of AI systems. However, little work has been done to characterize the flaws of image-based AIGC. In this work-in-progress, we analyzed 482 Reddit posts on various flaws of AIGC experienced by AIGC tool users. We found four themes describing the flaws of AIGC—logical fallacy, AI surrealism, misinformation, and cultural bias. We compare the results with the existing text-based AIGC framework on errors to discover unique flaws that image-based AIGC creates. We discuss implications toward a framework to describe, understand, and interpret flaws in AIGC in the broader context of understanding our social-technical world.

CCS Concepts

• **Human-centered computing** → HCI design and evaluation methods; Interaction design theory, concepts and paradigms; • **Computing methodologies** → Artificial intelligence; Discourse, dialogue and pragmatics; • **Hardware** → Emerging technologies.

Keywords

Artificial Intelligence, AI-generated images, AI-generated content, hallucination, AI bias and fairness, Chatbot, large language models, Human-AI Interaction

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1 Introduction

In 2022, Kalluri, a graduate student in artificial intelligence (AI) at Stanford University, observed a troublesome issue in image-generating AI programs [3]. When Kalluri prompted a photo of an

American man and his house, the program generated a pale-skinned person in front of a large, colonial-style home. When prompted ‘to a photo of an African man and his fancy house’, it generated an image of a dark-skinned person in front of a simple mud house despite the word ‘fancy’. This case was introduced in a news article published in Nature in March 2024 to illustrate that AI-generated content (AIGC) tools continue to amplify biases and stereotypes in proportions that are much greater than the demographic reality—as described by amplified stereotypes [10]. Images on the internet, including those from media outlets, global health organizations, and Wikipedia often present biased representations of gender, race, and nationality [16, 55]. These existing biases in training data will further exacerbate the stereotypes as AIGC continues to spread, populating internet websites and various marketing prints.

Researchers extensively investigated the types of harms, biases, and fairness in AI-generated text [18, 30, 38]. Numerous classification systems describing the different types of biases have been introduced [14, 22]. These commonly refer to concepts around how the text-based AIGC outputs factual, misleading, and discriminating errors [59] as well as logical, reasoning, math, and syntactic-related failures [9]. Other terms describing distorted and erroneous information included ‘fabrication’, ‘hoax’, ‘rumors’, ‘clickbait’, ‘manipulation’, ‘misappropriation’, and the list goes on [28]. Sun et al. [48] then consolidated the literature into a comprehensive classification system to place systematic, consistent labels upon distorted information.

Such standardization of language describing flaws and failures of AIGC can help researchers investigate further the causes and scenarios of AIGC errors and the field can analyze the erroneous information across different contexts through standardized naming convention. Having understandable language can also be used as a cautionary tale for individuals to approach AIGC tools with balanced perspectives.

However, despite the pressing issue that image-based AIGC brings in proliferating bias and social harms, no work has been done to date in developing standardized language to describe errors, biases, and failures of image-based AIGC. In this work-in-progress, we investigate the types of flaws that image-based AIGC generates and develop an initial set of vocabularies that uniquely illustrate those flaws. To address this gap, We collected 482 cases of image-based AIGC posted by Reddit users, who shared the content as ‘AI fails’ about their original intended request. We used open coding and thematic analysis to develop the vocabularies that describe the different types of flaws observed in the shared AIGC. We then compared our results with the existing standardized framework describing distorted information of text-based AIGC to understand what flaws are unique to image-based AIGC, and how we might further build up our standardized language around AIGC flaws

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and related problems that AIGC generates in propagating harmful information.

2 Related Works

Multimodal Large Language Model (MLLM) is a specialized term used to formally refer to the LLM-based model with the ability to receive, reason, and output with multimodal information. MLLM allows for text content creation grounded in text prompts and image [51], video [4, 20], and audio [35]. MLLM allows for not only input but also output in different modalities [26]. These advancing models significantly bolstered the advancement of multimodal AIGC, which has emerged as an innovative paradigm in content creation, capitalizing on automating the production of content on the internet. Over the years, AIGC has made significant progress in being applied to education, media, healthcare, finance, and entertainment [12]. Applying AIGC has helped to address critical societal challenges, such as resource scarcity and complex procedures in clinical contexts—working around diagnostic imaging, omics data, time-series, rich patient data, and other multi-modal clinical data [2].

With the increased use of AI systems and applications in our everyday lives, bias and fairness have gained significant attention among researchers designing and developing such systems [32]. These AI systems are often deployed in sensitive environments, aiding in making important and life-changing decisions. AI decisions should be free from any discriminatory behavior toward certain groups or populations. To better understand, communicate, and investigate fairness and related conceptualizations, researchers created taxonomy, classifications, and categories that characterize observed as well as hypothesized biases, potential harm, and fairness [1, 46, 47, 53]. For instance, chatGPT outputs have shown to demonstrate cases of overfitting, logic error, rational error, mathematical errors, unfounded fabrication errors regarding factual correctness, and text-based output errors [48]. Other studies found explicit and implicit biases of gender, race, and sexual orientation [23, 31]. General cultural overgeneralizations have been found rampant in the LLM outputs [19, 29, 49]. To mitigate cultural and contextual bias, researchers developed refined datasets for pertaining in which either remove harmful rhetoric or purposely include data from a wider variety of cultural and contextual sources [8, 57].

AI-based image recognition and detection software also face the challenges of bias and hallucination [44]. Researchers characterized the ‘hallucinations’ [43] found in Foundation Models [58], such as MLLMs. Researchers categorized ‘multimodal hallucinations’ into three types [6]: (1) ‘Object Hallucination’ is the most basic form, where models incorrectly claim the existence of certain objects in the image; (2) ‘Attribute Hallucination’ means describing the attributes of certain objects in a wrong way, e.g. failure to identify a dog’s color correctly; (3) ‘Relationship Hallucination’ is a more complex type and is also based on the existence of objects. It refers to false descriptions of relationships between objects, such as relative positions and interactions. There are also other characterizations [41], such as ‘contextual disconnection’, where the output produced is inconsistent with different modalities, and ‘semantic distortion’, in which the meaning of input has somehow been misinterpreted.

‘Content hallucination’ describes how features developed are unusual or absent. Factual inaccuracies are also problems in MLLMs [40], specifically on object counting and special relation understanding of the image outputs. We also observe trends of racial and gender bias in professional associations where the machine groups certain skin colors and genders to particular occupations [15].

While these definitions provide a broad overall understanding of the types of errors that occur, the term AI hallucination gives agency to the machine, anthropomorphizing it, leading to an exaggeration and misrepresentation of AI capabilities while simultaneously distorting the moral judgments of AI [7]. Bringing life to the technology makes users more forgiving of mistakes, relieving responsibility for the error [37]. The term also forces us to accept the good enough nature of technology production—the notion that developers should release good enough technology and updating over time. When working with LLMs, however, such ‘fix post-release’ practices can generate severe consequences, rendering the technology incapable of many of its assigned tasks [7].

As such, a myriad of literature attempted to define and classify bias, fairness, and harm. Often, these definitions can be overly broad or ill-defined. Researchers recognized such limitations and attempted to consolidate dispersed notions around bias and fairness into a unified framework. For instance, Sun et al. [48] systematically consolidated 75 distorted information types reported by scholars and developed a classification scheme describing distorted information produced by AI. This work, however, was primarily on text-based AIGC, and no special attention has been paid to image-based AIGC and their erroneous outputs. Below, we discuss how we attempted to address this gap in the literature.

3 Methods

This WIP answers the following research questions: (RQ1) *What are the different types of flaws in image-based AIGC?* (RQ2) *What are the common and unique characteristics of the flaws in image-based AIGC compared to that of text-based AIGC?* To answer these questions, we conducted a content analysis of Reddit Forums and compared the results with an existing classification of the distorted information in text-based AIGC [48].

Data Collection. We collected 750 posts from five subreddits: two on complaints about the output of AI technology (*r/aiFails*, *r/aifreakout*), one on chatbot interactions (*r/chatgpt*), and two on AI-generated images (*r/aiart*, *r/aiimages*). These forums contained a total of around 9 million users. From each forum, we collected the first 150 posts in the order the posts appeared between 11/25/2024 and 12/12/2024. The rules of the forums explicitly state that moderators remove any posts that do not pertain to fails on AIGC or related topics. We further confirmed that the images we collect from the posts were AI generated based on the posts’ title and comments. Examples include a title named, “What an AI thinks the Mona Lisa looks like from memory”. Not all posts included prompts that generated failed examples of AIGC. Accordingly, we did not require prompts to be a part of the dataset for inclusion. We screened out the posts based on the following exclusion criteria: (1) does not contain image-based AIGC, (2) the image-based AIGC introduced in the post was meant to be abstract artwork, (3) non-AI-generated images, such as memes, complaints, a screenshot of



Figure 1: AIG images themed under ‘Logical Fallacy’.

text-based AIGC, and (4) the post contained non-English content. After the screening, a total of 482 images remained for analysis.

Data Analysis. Two coders then iteratively conducted open coding [17] of the 482 images, where the primary coder coded for every image, the description on observed complaints, flaws, failures, or complaints about AIGC posters directly shared in their posts. The coder also looked for improbability–unrealistic or harmful characteristic–of the images. The coder iteratively checked with the second coder on the appropriateness of the codes being generated. We then conducted the axial coding exercise to consolidate the codes, which led to 16 initial categories of codes. We conducted affinity diagramming on these 16 codes. During this process, the 16 codes were consolidated if they were redundant, refined the code labels to further delineate the differences across the codes. This process generated four larger themes with ten codes underneath. The final codebook (Appendix 1) contains full definitions and examples.

4 Results

We found four overall themes and ten codes that best encompass the different types of reported flaws applicable to image-based AIGC. The themes included: Logical Fallacy, AI Surrealism, Misinformation, and Cultural Bias. We go through each theme and the codes underneath.

4.1 Logical Fallacy

The analysis showed 242 out of 482 images (71.0%) exhibited the theme of *Logical Fallacy*. In this case, the AI tool generated content that was either missing or over-abundant in crucial components (e.g., Not Summing to a Whole), had two elements or components of the content cancel one another out logically (e.g., contradiction) or reused the same sub-image or phrase within the larger output (e.g., redundancy).

4.1.1 Not Summing to a Whole. 232 out of 482 images showed the AIGC tool produced an output that did not align with the conventional sum of components required for the requested visual image to be completed. For instance, Fig. 1(a) shows when the cake pieces did not add up to a whole cake. Or, we see extra components for the requested image. Examples include having extra limbs, fingers, and facial features. Fig.1(b). We also see a deficit of components, either because it is missing or not fully formed, creating either underdeveloped (e.g., an eye with a misshapen iris and a missing waterline–Fig.1(c)).

4.1.2 Contradiction. 5 out of 482 images showed actions that contradict users’ original request or a sub-topic of their request. For instance, an object or building that contains a component that



Figure 2: The AIG images shown are themed under ‘AI Surrealism’.

contradicts its purpose (e.g., a building with an unreachable door, Fig.1(d)) The pathway does not properly lead to the door of the stylized castle generated rendering it unusable even in a stylized form contradicting its purpose as a building; the pathway also seems to have no entrance contradicting its purpose as well.

4.1.3 Redundancy. 5 out of 482 images showed the same sub-image or phrase unnecessarily repeated in completing users’ requests. We call these cases ‘Redundancy’. For example, Fig. 1(e) shows how the same facial image was used for multiple human figures in the image. Similarly, Fig.1(f) shows five repeated images of chandeliers in one room concentrated in one area, when common sense is that chandeliers if there are multiple, would usually be dispersed throughout the room for efficient lighting.

4.2 AI Surrealism

170 out of 482 images (35.2%) exhibited the theme of *AI Surrealism*. We found there was a reoccurring theme of uncanny or non-photorealistic appearance. This appearance occurred under a few sub-conditions: the piece seemed to have elements at a different level of detail (e.g., unfinished); had overly detailed, uncharacteristic images of what a camera could capture or what exists in reality; or had an unrefined and dreamy appearance (e.g., smoothness).

4.2.1 Unfinished. 21 out of 482 images showed different sections of the images had different levels of detail, implying that the requested detail has not been applied to the full image. For instance, in an image of two cats side-by-side, one cat’s features are relatively well developed, while the other cat has no defined limbs or face (Fig.2(a)). We observe from these cases that AI is capable of developing a well-rendered image, but it does not apply the development to the entire image output.

4.2.2 Over-detail. 20 out of 482 images showed specific details that reoccur at an improbable frequency. A frequent case is how highlights of the face, hair, and clothing are excessive Fig 2(b). Figure 2.1 shows highlights shining on the statues in the back as well as money in the foreground that flatten the image and remove potential depth. Such flaws generate unrealistic output and a lack of foreground-background contrast.

4.2.3 Smoothness. 129 out of 482 images showed the foreground and background have the same level of smoothness, lacking natural camera grain or textured details that seem overly blended and blurred. For instance, a birthday cake frosting on the plate Fig. 1(a) lacks the usual grain seen in photographs. The newspaper, teacup, cake, and knife all exist with the same level of grain and clarity.

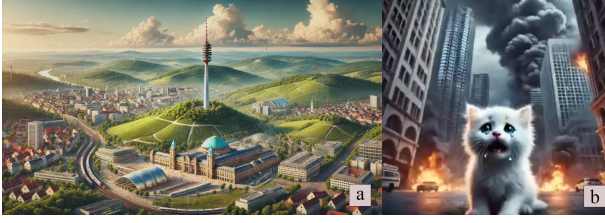


Figure 3: The AIG images are themed under ‘Misinformation’.

Other cases include hair and fabric (Fig. 2(c)), where individual strands are blended into one another and do not maintain singularity throughout their length like in the real world. And, the shirt in the figure lacks the usual fabric texture.

4.3 Misinformation

67 out of 482 images (13.9%) exhibited the theme of *Misinformation*, where the images partially or fully misrepresented a known reality or fact. Sub-codes of this theme included *factually untrue* and *improbable or unsafe*, which explained multiple aspects of misrepresentation of the truth and reality.

4.3.1 Factually Untrue. 34 out of 482 images contained some form of objectively untrue information, about users’ requests.

In these cases, the mistakes the tool makes are indubitably incorrect. For instance, in Fig.3(a), the user inputs a request for an image of Stuttgart, Germany, and the output does not provide a correct image of the city. The excessive number and location of hills as well as a tower appearing in this image is not representative of Stuttgart. The architectural forms were also inaccurate.

4.3.2 Improbable. 51 out of 482 images contained improbable content. These images contained content that is illogical, dangerous, or defying basic physics and biology. For instance, tears should fall on fur under gravity. However, Fig.3(b) shows the kitten’s teardrop swells and falls in a manner that does not overlap properly with the fur on its body.

4.4 Cultural Bias

35 out of 482 (7.3%) images exhibited the theme of *Cultural Bias*. In this case, the AIGC tool over or underused the common appropriate concepts related to cultural understanding [45]. In this case, the output either contains harmful or oversimplified rhetoric (e.g., *stereotypes*) or lacks common cultural understanding (e.g., *Lack of cultural understanding*).

4.4.1 Stereotypes. 11 out of 482 images contained harmful, oversimplified rhetoric commonly described as stereotypes. For instance, in Fig.4(a), the user prompted the tool to depict an image of a “17th-century British king eating watermelon”. The tool produced an output in which the clothing and accessories align with that of a British king of the 17th century, but it coated the eating watermelon with a racial stereotype and depicted the king as African American. This is a stereotype that is inappropriate and harmful content for the AI to be distributing as output for the user and a greater culture of content online.

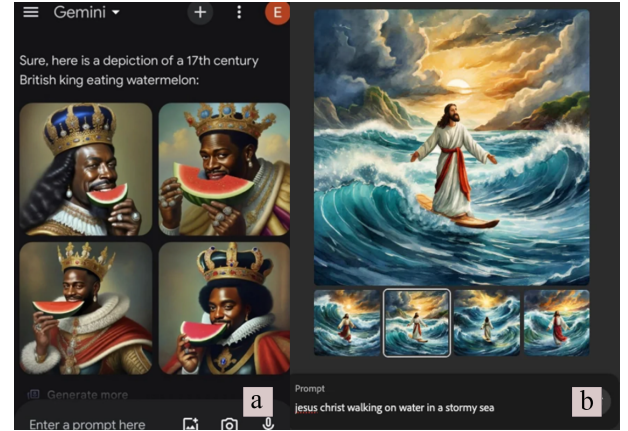


Figure 4: The AIG images are themed under ‘Cultural Bias’.

4.4.2 Lack of Cultural Understanding. 24 out of 482 images contained crude, inappropriate, or harmful suggestions with a lack of common cultural convention knowledge. For instance, in the case of biblical stories, the user prompted the tool to depict “Jesus Christ walking on water in a stormy sea” Fig. 4(b). The tool produced four images in which Christ is on a surfboard in a stormy sea. A common story in culture and on the internet is the story of Christ walking on water. This was not taken into account when generating the image. Instead, there is a common understanding of how a human could possibly be floating on the water—that it is not possible without a device, such as a surfboard.

4.5 Unique Flaws of Image-Based AIGC

Table 1 shows the comparisons made between our results on the themes and the codes on flaws observed about AIG images versus the existing bias framework on text-based AIGC by Sun et al. [48]. We found that, except the AI Surrealism theme (3 codes), all other themes had only one code that showed common flaws between AIG images versus text, while the rest of the 4 codes across the themes were uniquely observed in AIG images. For instance, under logical fallacy, the ‘contradiction’ code overlapped with Sun et al.’s ‘Contradictions’, which was defined as situations in which the generated content is self-contradictory or inconsistent [11]. Similarly, the ‘factually untrue’ code of AIG images overlapped in concept with a few codes from Sun et al.’s framework—‘Objective Fact Errors’, ‘Fake News’, ‘Pseudoscience’, and ‘Common Sense Mistakes’. These codes are commonly referred to as ‘unfounded fabrication’, where the AI generates claims based on the question, distorting news reports and creating scientifically dubious claims [11]. These cases also produced output contrary to common sense or generally accepted knowledge [13, 28, 34]. Additionally, the ‘stereotypes’ code of AIG images were similar to the ‘hate news’, ‘discrimination’, and ‘bias’ codes by Sun et al. The common theme here was that the generated content was intentionally derogatory to certain ethnic groups or the content reflected cultural biases based on race, gender, religion, ethnicity, social status, and cultural background [11, 13, 34]. As a result, the concepts illustrated through the codes—‘not summing to

Themes	Codes	Codes for Text-Based AIGC [48]
Logical Fallacy	Not Summing to a Whole	N/A
	Redundancy	N/A
	Contradiction	Contradictions
AI Surrealism	Unfinished	N/A
	Over Detail	N/A
	Smoothness	N/A
Misinformation	Factually Untrue	Objective Fact Errors, Fake News, Pseudoscience, Common Sense Mistakes
	Improbable	N/A
Cultural Bias	Stereotypes	Hate News, Discrimination, Bias
	Lack of Cultural Understanding	N/A

Table 1: This table shows which codes on image-based AIGC overlap or are unique to the AIG image flaws, compared to the existing bias framework on text-based AIGC [48].

a whole’, ‘redundancy’, ‘unfinished’, ‘over detail’, ‘smoothness’, ‘improbable’, and ‘lack of cultural understanding’ were newly observed flaws that AIG images have shown.

5 Discussion

Similar to the bias and failures observed in LLM [36], image-based AIGC has the potential to spread misinformation and perpetuate harmful content to users. Wilson uses Kant’s duty of truthfulness to argue that we have the moral obligation of self-honesty, which in turn is providing truthful information [42]. This truthful obligation can be brought into the scope of information technology in which we can examine Frankfurt’s analysis of the liar [21], which, in relation to the truth, is said: “A [liar]’ does not purposefully distort the truth, but rather is ‘disconnected from a concern with the truth’... ‘is a greater enemy of the truth than lies are’ because [liars] have no regard for the truth, their only aim is to make people believe them at any cost ” [21, 50, 54]. This is parallel to the goal of image-based AIGC and LLM, where the goal of the AI seems to provide the user with a convincing output that is not truthful [50]. Scientists and educators have an ethical obligation to tell the truth and communicate uncertainty, which is undermined by the assertive nature of LLMs. This is paralleled in image generation AI tools [39, 50].

With cultural bias, AIGC image outputs often contain stereotypes that can push false narratives about certain demographics and also provide outputs that are overfitted to a particular group of

people. Images are powerful tools to perpetuate ideas and persuade people compared to other less visual forms of media [33]. Accordingly, potential harms observed in AIG images need to be critically examined. In a case study analyzing images developed for a website relating to education, researchers asked various image-based AIGC tools to construct a diverse classroom setting with a black female teacher. Every image was deemed unusable for their website, and often, the teacher figure was also present with a white male or female seemingly in authority [56].

Existing frameworks on erroneous information, bias, failure, or flaws of AIGC and Foundation Models have focused on the spread of misinformation and logical flaws, such as contradicting information or fact errors, fake news, or common sense mistakes. Also, discrimination, bias, and hate news are more well-known, explicit forms of stereotypes being perpetuated in AIGC. We observed all of these flaws existing in visual forms in the AIG images. At the same time, we observed flaws that are unique to AIG images, such as how sub-objects of a concept are either missing or added too much—e.g., too many table legs, lack of tires for a four-wheel car. The identified themes of this work provide unique perspectives on how multi-modal AIGC can continue to be evaluated and investigated for its flaws and mitigation strategies. Further, our work contributes to CHI as a stepping stone to further expanding understanding around image-based AIGC and the potential harms they may perpetuate through various media.

These harms impact the overall functionality of the AI systems, and the current guidelines around trust and bias of AI systems are focused on building stakeholders’ trust in AI products [27]. This trustworthiness is obtained when the technology can fulfill the public’s expectations [25, 27]. The ideal of democratizing AI systems for public use is reliant on their functionality, which is currently failing [5]. When culturally harmful or inaccurate information can be output by the system, just as in their text-based LLM counterparts, malicious users can leverage these flaws for their benefit [52]. LLM models have been exploited to spread unsafe information, such as how to construct weaponry or convince users to act on social thoughts [24]. Our work in discovering fails of image-based AIGC brings a viewpoint on how image-based AIGC can generate new types of broader risks and harms that can more effectively impact users’ mental and physical safety.

5.1 Limitations

The scope of our work is limited due to self-selection bias—the images we collected were not representative of all image-based AIGC. The data were based on users who voluntarily shared flawed AIGC images. Thus, the themes reported here are not representative of the broader contexts. However, investigating what users have already highlighted as their experiences around seeing flaws AI produces with images has allowed us to conduct focused research that can serve as a stepping stone to more systematized, generalizable findings. Future work should build on this work to develop a refined framework that describes the AI flaws and the implications for the social and technical world. We should include more diverse dataset, such as social media sites, to further refine the framework. We can also systematically prompt and replicate the flawed images to validate the results.

6 Conclusion

We found four generalized themes: misinformation, Logical Fallacy, AI surrealism, and Cultural Bias, each consisting of two to three sub-codes defining refined and specific case scenarios observed in the dataset. The results partially overlapped with the existing text-based AIGC, but unique and novel themes pertaining to images only emerged. Our work contributes to building a stepping stone for the ACM CHI community in discussing a unified, emerging framework regarding image-based AIGC and its flaws, biases, and fairness.

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